The Zeta Function of the T^4/Z_3 Torus:

Orbifold Projection, Dirichlet L-Functions and Sector Pairing of the GF(27) Classes

Johann Pascher
ORCID 0009-0000-6518-4064

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Abstract

The Galois factorisation classes of Doc. 342 and the harmonic series on the T^4 torus of Doc. 159 together yield a precise statement about the zeta function of the FFGFT spacetime.

Four results: **(A)** The Z_3 orbifold projection of T^4/Z_3 transforms the full Riemann zeta into a Dirichlet L-function: $\zeta_{T^4/\mathbb{Z}_3}(s) = L(s,\chi_0) = (1-3^{-s})\zeta(s)$ **[B]**. **(B)** The Euler product of this L-function is hierarchically ordered by the entry depth $k = \mathrm{ord}_p(3)$; the physically visible primes $\{2,3,5,7,11,13\}$ are exactly those with $k \le 6$ **[B]**. **(C)** The orbifold projection generates additional zeros at $s = 2\pi ik/\ln 3$ ($k \in \mathbb{Z} \setminus \{0\}$) on the imaginary axis $\mathrm{Re}(s) = 0$; these are the algebraic signature of the Z_3 sector separation **[B]**. **(D)** The four Frobenius conjugacy classes of primitive elements in $\mathrm{GF}(27)^*$ (Doc. 342, Theorem A) are paired into *two* classes by the FFGFT sector pairing $k \leftrightarrow -k$ (Doc. 336; on $\mathrm{GF}(27)^*$: $x \mapsto x^{-1}$), $f_1 \leftrightarrow f_4$ and $f_2 \leftrightarrow f_3$; negation $x \mapsto -x = x^{13}x$ couples each primitive class to exactly one order-13 class **[B]**. The Galois character of $\mathrm{Gal}(\mathrm{GF}(27)/\mathrm{GF}(3)) \cong \mathbb{Z}_3$ does not distinguish the classes; the distinction lies in $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$. Whether exactly one second $\mathrm{GF}(27)^*$ layer besides the lepton layer is physically realised remains open **[S]**.

Two further results: **(E)** The harmonic primes $\{5,7,11,13\}$ are uniquely forced by the prime factorisations of $3^k - 1$ for $k = 3,4,5,6$; $p = 13$ is the worldwide unique prime with $\mathrm{ord}_p(3) = 3$ **[B]**. **(F)** The Euler factor threshold at $p^* \approx 9.76$ ($s = 2$) marks the 7-limit exactly; from $p_{\max} \approx 30$ the Galois product density lies at chance level; the Weyl obstruction (Doc. 316) shows that the limit lies in the object — no quantum computer of any size circumvents it **[B]**.

Check scripts: `pruef_343_zeta_galois.py`, `pruef_343b_klassen_symmetrie.py`, `pruef_344_primzahlmuster.py`.

.1 Status markers and starting point

The following status markers are used throughout:

- **[K]** *Confirmed* — fully proved from ξ or a corpus-established derivation; all check-script assertions passed.
- **[B]** *Established* — numerically secured; analytical proof present or trivial.
- **[S]** *Speculative* — indication or conjecture, not yet fully proved.

Doc. 159 shows that the spectral zeta function of the Laplace operator on a torus T^d is the Epstein zeta function, whose one-dimensional special case is the Riemann zeta $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Doc. 342 shows that primes appear in the GF(3^k) hierarchy of the T^4/Z_3 orbifold ordered by entry depth $k = \mathrm{ord}_p(3)$. This document combines both: what does the GF(3^k) hierarchy imply for the zeta function itself?

.2 Theorem A: Orbifold projection and Dirichlet L-function

Decomposition of the zeta by Z_3 classes

On T^4/Z_3 the allowed modes $n \in \mathbb{Z}^4$ are structured by Z_3 invariance. In the one-dimensional analogue $\zeta(s)$ splits into three partial sums by $n \bmod 3$ **[B]**:

$$\zeta(s) = \underbrace{\sum_{\substack{n=1\\3|n}}^{\infty} \frac{1}{n^s}}_{=3^{-s}\zeta(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv 1(3)}}^{\infty} \frac{1}{n^s}}_{L_1(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv 2(3)}}^{\infty} \frac{1}{n^s}}_{L_2(s)} \tag{1}$$

The fixed-point sector ($n \equiv 0$) contributes exactly $3^{-s}\zeta(s)$; the twist sectors ($n \equiv \pm 1$) together $(1 - 3^{-s})\zeta(s)$.

Principal character χ_0 mod 3

The Dirichlet character χ_0 mod 3 with $\chi_0(n) = 0$ if $3 \mid n$, otherwise $\chi_0(n) = 1$, gives **[B]**:

$$L(s, \chi_0) = \sum_{\mathrm{gcd}(n,3)=1} \frac{1}{n^s} = (1 - 3^{-s})\,\zeta(s) = \prod_{p\neq 3} \frac{1}{1 - p^{-s}} \tag{2}$$

This is the **spectral zeta function of the T^4/Z_3 torus**: the characteristic prime $p = 3$ does not appear as an Euler factor — it is the geometric foundation of the orbifold and stands outside the prime hierarchy.

Non-trivial character χ_1 mod 3

The character χ_1 mod 3 with $\chi_1(1) = +1$, $\chi_1(2) = -1$, $\chi_1(0) = 0$ measures the *asymmetry* of the twist sectors **[B]**:

$$L(s, \chi_1) = \sum_{\substack{n=1\\n\equiv 1(3)}}^{\infty} \frac{1}{n^s} - \sum_{\substack{n=1\\n\equiv 2(3)}}^{\infty} \frac{1}{n^s} \tag{3}$$

with the exact value (Dirichlet 1837): $L(1, \chi_1) = \pi/(3\sqrt{3})$.

Physical meaning **[S]**: $L(s, \chi_1)$ measures the difference between winding numbers $w = +1$ and $w = -1$ on the torus, i.e. the Z_3 sector asymmetry. Whether $L(s, \chi_1)$ has a direct physical observable in FFGFT (particle–antiparticle asymmetry in the twisted sector?) remains open **[S]**.

.3 Theorem B: Hierarchical Euler product by $k = \mathrm{ord}_p(3)$

The primes in $L(s, \chi_0) = \prod_{p\neq 3}(1 - p^{-s})^{-1}$ are not equivalent. By the entry-depth rule $k = \mathrm{ord}_p(3)$ from Doc. 342, Theorem B, the Euler product is hierarchically ordered **[B]**:

$$L(s, \chi_0) = \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \frac{1}{1 - p^{-s}} \tag{4}$$

Table 1: Hierarchy of Euler factors by $k = \mathrm{ord}_p(3)$

k	new prime(s)	Euler factor at $s = 2$	harmonics	FFGFT role
1	2	$4/3 = 1.333$	octave	Z_2 fixed point, massive sector
3	13	≈ 1.006	thirteenth	GF$(27)^*$, leptons
4	5	≈ 1.042	major third	factor in $(m_\mu/m_e)^2$
5	11	≈ 1.008	eleventh	neutrinos (Doc. 340)
6	7	≈ 1.021	seventh	top quark (Doc. 289)
≥ 7	$\geq 41, 1093, \ldots$	$\approx 1 + p^{-2}$	no small harmonics	suppressed

Fractal suppression of higher Euler factors

On the fractal T^4 torus (Doc. 159) high-mode contributions are suppressed by the damping factor $\tau^{-(1+\xi)k}$. This transfers to the Euler factors; the *weighted* Euler product reads

$$\zeta^{\mathrm{frac}}_{T^4/\mathbb{Z}_3}(s) \;\cong\; \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \left(\frac{1}{1 - p^{-s}}\right)^{\tau^{-k(1+\xi)}} \tag{5}$$

Since $\tau^{-k(1+\xi)} \to 0$ as $k \to \infty$, each Euler factor is pushed toward 1 for large k: primes with $\mathrm{ord}_p(3) \geq 7$ make no physical contribution **[B]**. This is Theorem D of Doc. 342 in the language of the zeta function.

.4 Theorem C: New zeros from the orbifold projection

Zeros of $(1 - 3^{-s})$

The factor $(1 - 3^{-s})$ in $L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$ has zeros at **[B]**:

$$3^{-s} = 1 \quad\Longleftrightarrow\quad s = \frac{2\pi i\, k}{\ln 3}, \qquad k \in \mathbb{Z} \setminus \{0\} \tag{6}$$

These lie on $\mathrm{Re}(s) = 0$ (imaginary axis), *not* on the critical line $\mathrm{Re}(s) = 1/2$ of the Riemann hypothesis. They are absent from the Riemann zeta spectrum; they arise solely from the Z_3 orbifold projection.

The zero spacing $2\pi/\ln 3 \approx 5.72$ encodes the periodicity of the Z_3 lattice structure. A resonance test of the first eight known Riemann zeros γ_n against this unit shows no significant coincidence (deviations 0.15–0.47) **[B]**: the full Riemann zeta encodes the distribution of *all* primes, not only the Z_3-compatible ones.

Complete zero structure of the orbifold zeta

$L(s, \chi_0)$ has three types of zeros:

- **Trivial zeros of $\zeta(s)$**: at $s = -2, -4, -6, \dots$
- **Non-trivial Riemann zeros**: on $\mathrm{Re}(s) = 1/2$ (assumed), unchanged
- **Orbifold zeros**: at $s = 2\pi i k/\ln 3$, $k \neq 0$ — new, generated by the Z_3 projection **[B]**

.5 Theorem D: The four classes under sector pairing

The Galois character does not distinguish the classes

The Galois group Gal(GF(27)/GF(3)) $= \langle F\rangle \cong \mathbb{Z}_3$ with $F\colon\ x \mapsto x^3$ has three characters $\chi_j(F) = \omega^j$, $j = 0, 1, 2$. All four primitive classes $f_1, \dots, f_4$ of Doc. 342 are *regular* F-orbits of length 3: F acts on each as a 3-cycle, and the associated character is the same for all four. The Galois character therefore does not separate the classes **[B]**.

The distinction lies in the discrete logarithm: with a generator x of GF$(27)^*$, each class is an orbit $\{x^k, x^{3k}, x^{9k}\}$ with gcd$(k, 26) = 1$, i.e. a coset of $\langle 3\rangle = \{1, 3, 9\}$ in $(\mathbb{Z}/26)^* \cong \mathbb{Z}_{12}$. There are $12/3 = 4$ cosets **[B]**:

Table 2: The four primitive classes as cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$ (basis $x^3 + 2x + 1 = 0$)

Class	Minimal polynomial	Exponents k	Role
f_2	$x^3 + 2x + 1$	$\{1, 3, 9\}$	Doc. 341, lepton layer
f_3	$x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	$= f_2^{-1}$
f_1	$x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	
f_4	$x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	$= f_1^{-1}$

The quotient $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$ is cyclic; inversion $k \mapsto -k$ is its element of order 2.

Sector pairing reduces four to two

Doc. 336 identifies the FFGFT sector pairing $k \leftrightarrow -k$ with the Frobenius automorphism; at the level of elements of GF$(27)^*$ it is inversion $x \mapsto x^{-1} = x^{25}$. It permutes the classes as **[B]**

$$f_1 \longleftrightarrow f_4, \qquad f_2 \longleftrightarrow f_3. \tag{7}$$

If sector pairing is a symmetry of the mass layer — as it is in the corpus — there are not four but **two** physically distinguishable GF$(27)^*$ layers: $\{f_2, f_3\}$ (leptons, Doc. 341) and $\{f_1, f_4\}$.

Negation couples order 26 to order 13

Since $-1 = x^{13}$ in GF(27), the map $x \mapsto -x$ sends every element of order 26 to one of order 13 and vice versa. The four order-13 classes $g_1, \dots, g_4$ of Doc. 342 are therefore not an independent structure but the sign partners of the primitive classes **[B]**:

$$f_1 \leftrightarrow g_1, \quad f_2 \leftrightarrow g_3, \quad f_3 \leftrightarrow g_2, \quad f_4 \leftrightarrow g_4. \tag{8}$$

The candidate "sub-lepton structure" from the order-13 polynomials, left open in Doc. 342, is thereby eliminated.

Open question

The reduction $4 \to 2$ sharpens the question of Doc. 342: not "do f_2, f_3, f_4 carry further layers", but "is there exactly one second GF$(27)^*$ layer $\{f_1, f_4\}$ besides the lepton layer $\{f_2, f_3\}$, and which particles carry it?" This remains open **[S]**.

.6 Theorem E: Prime patterns in the GF(3^k) hierarchy

The harmonic primes are uniquely forced

A prime p satisfies $\mathrm{ord}_p(3) = k$ if and only if p divides $3^k - 1$ and $p \nmid 3^j - 1$ for all $j < k$. The prime factorisations of the first relevant values give **[B]**:

Table 3: Prime factors of $3^k - 1$ and the forced physical primes

k	$3^k - 1$	prime factors	phys. prime	note
1	2	$\{2\}$	2	octave
3	26	$\{2, 13\}$	**13**	unique prime with $\mathrm{ord}_p(3) = 3$ worldwide
4	80	$\{2, 5\}$	**5**	only non-2 factor
5	242	$\{2, 11\}$	**11**	only non-2 factor
6	728	$\{2, 7, 13\}$	**7**	(13 has $\mathrm{ord}_{13}(3) = 3$, not 6)

The strongest result concerns $p = 13$ **[B]**: since $3^3 - 1 = 26 = 2 \cdot 13$ and $\mathrm{ord}_2(3) = 1$, $p = 13$ is the *unique* prime worldwide with $\mathrm{ord}_p(3) = 3$. There is no other, and there can be none. The physical primes $\{5, 7, 11, 13\}$ are not selected by harmonic considerations, but are completely and uniquely forced by the prime-factor structure of $3^k - 1$ for $k = 3, 4, 5, 6$.

Necessary congruence condition

From $\mathrm{ord}_p(3) = k$ with $3 \mid k$ it follows that $3 \mid (p - 1)$, i.e. $p \equiv 1 \pmod 3$ **[B]**. In particular $p = 13$ (at $k = 3$) and $p = 7$ (at $k = 6$) are both $\equiv 1 \pmod 3$ — exactly the Z_3-compatible primes of the corpus (Doc. 342, Theorem B).

.7 Theorem F: Limits of distinguishability and computational cost

Exact threshold: Euler factor vs. fractal correction

The Euler factor $1/(1 - p^{-s})$ exceeds the fractal correction $1.05\,\%$ (Doc. 338, bare to measured) exactly for $p < p^*$ with $p^* = (0.0105)^{-1/s}$ **[B]**:

Table 4: Exact threshold p^* by value of s

s	p^*	primes $\leq p^*$	harmonic limit	Euler factor $\geq 1.05\,\%$
1	95.2	$\{2, 3, 5, \ldots, 89\}$	—	all small primes
2	9.76	$\{2, 3, 5, 7\}$	7-limit	octave, fifth, third, seventh
3	4.57	$\{2, 3\}$	3-limit	octave and fifth only

For $s = 2$ (reference value of the spectral zeta) the threshold is $p^* \approx 9.76$: primes $\{2, 5, 7\}$ have Euler factors exceeding the fractal correction; $p = 11$ and all larger primes fall below it. This is exactly the harmonic 7-limit. $p = 11$ appears in the corpus (neutrinos, Doc. 340) via the structural formula $\Delta m^2 = (11^2 - 1)\, m_\nu^2$, not as a Galois product hit.

Galois products vs. random products: hit rate

Forming all products of up to three Galois building blocks with exponents $\{-1, 1, 2\}$ for $p \leq p_{\text{max}}$, the product set grows rapidly and the hit rate for arbitrary targets reaches chance level **[B]**:

Table 5: Hit rate for physical vs. random targets at $\pm 0.5\,\%$

p_{max}	values	43200	m_μ/m_e	random avg
13	1334	2	0	0.92
30	5802	3	2	3.60
50	8946	3	2	5.33
100	13398	4	3	7.66
300	24992	8	12	14.15

From $p_{\text{max}} \approx 30$ the hit rate lies at chance level ($\pm 0.5\,\%$); at the strict threshold $\pm 0.05\,\%$ already from $p_{\text{max}} \approx 13$. The identity $43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$ (Doc. 338) uses only building blocks with $p \leq 5$ and is derived *geometrically* from ξ, not found by product search.

Structural computational limit: Weyl obstruction and quantum computers

The distinguishability threshold at $p_{\text{max}} \approx 13$ is a *physical* limit (fractal correction). Independently of this, the corpus contains a sharper *mathematical* limit that applies to every technology **[B]**:

Weyl obstruction, Doc. 316

There is no compact Riemannian manifold — of any dimension, curvature, topology, or orbifold structure — whose Laplace spectrum consists of the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

From Doc. 147, an n-qubit register is a compact state space with eigenvalue density $N(\lambda) \sim \lambda^{d/2}$; arithmetic counting densities grow as $T \ln T$. A finite state space cannot fully carry such a spectrum, *independently of the register size* **[B]**.

Doc. 075 names the consequence for Shor's algorithm:

Method	Limit	Type
Trial division	$O(\sqrt{n})$	computational
Pollard ρ	$O(n^{1/4})$	computational
Shor's QFT	resource limit	Weyl obstruction

The central statement of Doc. 316 holds technology-independently:

> *The limit lies not in the tool but in the object.* Transformation (Fourier, QFT, optics) projects and generates no information. Relaxation finds what is already in the potential. Both presuppose that the sought structure is already present.

Physical limit ($p^* \approx 9.76$, Theorem F.1) and mathematical limit (Weyl obstruction, Theorem F.3) thus appear in the same picture: beyond a certain point the prime structure is hidden in the noise, and no tool — including an arbitrarily large quantum computer — retrieves it.

.8 Summary

Table 6: Results and status

Theorem	Content	Status
A	$\zeta_{T^4/\mathbb{Z}_3}(s) = (1-3^{-s})\zeta(s) = L(s,\chi_0)$	**[B]**
A′	$L(s,\chi_1)$ measures twist-sector asymmetry	**[S]**
B	Euler product hierarchical by $k = \mathrm{ord}_p(3)$	**[B]**
B′	Fractal damping suppresses Euler factors $k \geq 7$	**[B]**
C	Orbifold zeros at $s = 2\pi ik/\ln 3$ on $\mathrm{Re}(s) = 0$	**[B]**
C′	No resonance of Riemann zeros with Z_3 unit	**[B]**
D	Galois character does not separate; classes $= (\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$	**[B]**
D′	Sector pairing $k \leftrightarrow -k$: $f_1 \leftrightarrow f_4$, $f_2 \leftrightarrow f_3$ — 4 classes → 2	**[B]**
D″	Negation couples order 26 to 13; no sub-lepton layer	**[B]**
D‴	Second layer $\{f_1, f_4\}$ physically realised?	**[S]**
E	$p = 13$ unique prime with $\mathrm{ord}_p(3) = 3$ worldwide; primes forced by $3^k - 1$	**[B]**
E′	$\mathrm{ord}_p(3) = k$, $3\mid k \Rightarrow p \equiv 1 \pmod 3$ (Z_3-compatible)	**[B]**
F	Euler factor threshold $p^* \approx 9.76$ ($s = 2$) = 7-limit boundary	**[B]**
F′	Product density: chance level from $p_{\max} \approx 30$ ($\pm 0.5\,\%$)	**[B]**
F″	Weyl obstruction: limit lies in the object, not the tool (Doc. 316)	**[B]**

The central statement: the full Riemann zeta $\zeta(s)$ belongs to the flat $\mathbb{R}^4$ spectrum. The zeta of the T^4/Z_3 orbifold is $L(s,\chi_0) = (1-3^{-s})\zeta(s)$ — a filtered version in which the 3-Euler factor is separated out and the primes are weighted by the Galois hierarchy $k = \mathrm{ord}_p(3)$. This is the zeta-function form of the entry rule of Doc. 342.

Check scripts

Directory `2/python/Dok343_Skripte/`, wrapper `run_all_343.sh`:

- `pruef_343_zeta_galois.py` — Theorems A–D; partial zeta decomposition, L-functions, zeros, resonance test, weighted Euler product.
- `pruef_343b_klassen_symmetrie.py` — Theorem D; explicit Frobenius classes in $\mathrm{GF}(27)^*$, action of inversion and negation, cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$, exact values of $L(s,\chi_1)$.
- `pruef_344_primzahlmuster.py` — Theorems E–F; patterns in $\mathrm{ord}_p(3)$, exact threshold p^*, Galois vs. random hit rate, Euler factors.

All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All numerical computations are verified by Python check scripts. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

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